2025–2026 春夏学期数据结构基础期中考
真题整理（李松）

一、判断题 (True or False)

R1-1. *(3 分)* If a complete binary tree has $n+1$ nodes in the last level and n nodes in the second last level, then the total number of nodes in the tree must be $3n$. (T/F)

R1-2. *(2 分)* A segment tree can be used to find the greatest common divisor (GCD) of any index range. (T/F)

R1-3. *(2 分)* If a connected graph G has no topological sorting sequence, then graph G must have a cycle. (T/F)

R1-4. *(2 分)* In Union/Find algorithm, if Unions are done by size, the depth of any node must be no more than $N/2$, but not $O(logN)$. (T/F)

R1-5. *(2 分)* $N!$ is $O(N^N)$. (T/F)

R1-6. *(2 分)* An algorithm may or may not produce output, but each algorithm is expected to accept at least one input. (T/F)

R1-7. *(2 分)* In a singly linked list of N nodes, the time complexities for query and insertion are $O(1)$ and $O(N)$, respectively. (T/F)

R1-8. *(2 分)* An algorithm to check for balancing symbols in an expression uses a stack to store the symbols. (T/F)

二、单选题 (Multiple Choice)

R2-1. *(5 分)* The array representation of the disjoint sets is given by S = { -4, 0, 0, 2, -3, 4, 5, -1 }, the elements are numbered from 0 to 7. Which operations can not turn the array into S = { -5, 0, 0, 0, -3, 4, 5, 0 }?

A. Find(5); Find(3); Union(0,7);

B. Union(find(2),7); Find(3);

C. Find(6); Find(3);Union(0,7);

D. Union(Find(3),7)

R2-2. *(5 分)* Given a directed graph of 8 vertices and 12 edges. Which one of the following statements is true?

A. The sum of in-degrees of all the vertices is 12.

B. The sum of degrees of all the vertices is 12.

C. There are totally 24 nodes in the adjacency list representation of the graph.

D. There are 24 non-zero elements in the adjacency matrix representation of the graph.

R2-3. *(5 分)* For a binary tree, if its post-order travel sequence is { 1, 3, 2, 6, 5, 7, 4 }, and its in-order travel sequence is { 1, 2, 3, 4, 5, 6, 7 }, then which of the following statement is TRUE?

A. This is a complete binary tree.

B. 6 is the parent of 5.

C. This is a binary search tree.

D. 3 is the parent of 2.

R2-4. *(5 分)* Which of the following statements is correct regarding a binary search tree storing unique integers?

A. The integers 8 and 9 may be at the same level.

B. If 8 and 10 are at the same level, then 9 must be their parent node.

C. If 8 and 10 are at the same level, then 9 must not be their parent node.

D. None.

R2-5. *(5 分)* Which of the following statements regarding a standard Segment Tree constructed for an array of n elements is TRUE?

A. The time complexity of building the Segment Tree from the initial array is $\Theta(n \log n)$.

B. In the worst case, both a single point update and a single range sum query take $O(\log n)$ time.

C. A Segment Tree can only be used for range sum queries and cannot support range maximum or minimum queries (RMQ).

D. The total number of nodes required to store a Segment Tree for an array of size n is always exactly $2n$.

R2-6. *(5 分)* Given an array of integers { 7, 3, 6, 2, 5, 1, 4 }. Build a min-heap using the linear algorithm. Then which one is the list of nodes along a path from the root to a leaf node of the resulting heap?

A. 1, 2, 5

B. 1, 2, 6

C. 1, 5, 3

D. 1, 3, 6

R2-7. *(5 分)* For the following function (where $n > 0$)

```
int func ( int n )
{   int i = 1, sum = 0;
    while ( sum < n )  { sum += i; i *= 2; }
    return i;
}
```

the most accurate time complexity bound is:

A. $O(\log n)$

B. $O(n)$

C. $O(n \log n)$

D. $O(2^n)$

R2-8. *(5 分)* Among the following operations for a sorted sequencial list (of length n), which one has the average time complexity of $O(1)$?

A. Find the element with a given key value

B. Insert the element with a given key value

C. Delete the i-th $(1 \le i \le n)$ element

D. Visit the i-th $(1 \le i \le n)$ element

R2-9. *(5 分)* Which of the following functions is $\Theta(n!)$?

A. $(n+1)!$

B. $\left(\lfloor \frac{n}{2} \rfloor !\right)^2$

C. $n! + 2^n$

D. n^n

R2-10. *(5 分)* For a non-empty doubly linked circular list, with `h` and `t` pointing to its head and tail nodes, respectively, the FALSE statement is:

A. `t->next == h`

B. `h->pre == t`

C. `t->pre->next == t`

D. `h->next == t`

R2-11. *(5 分)* Given the following postfix expression: "5 3 + 6 2 - *", what is its value?

A. 10

B. 16

C. 24

D. 32

R2-12. *(5 分)* The following list is a series of operations for a stack:

- push 1
- push 2
- push 3
- pop
- pop
- push 4
- push 5
- pop
- push 6
- push 7
- pop
- pop
- pop
- push 8
- pop

- push 9
- pop
- pop

Which is the correct pop-up sequence?

A. 1 2 3 4 5 6 7 8 9

B. 9 8 7 6 5 4 3 2 1

C. 3 2 5 7 6 4 8 9 1

D. 3 2 5 7 4 6 9 8 1

R2-13. *(5 分)* A circular queue is stored in A$[0, 1, \cdots, n-1]$,its head and tail pointer are f, r.How many elements does the queue have?

A. $r - f - 1$

B. $r - f$

C. $(r - f)\%n + 1$

D. $(r - f + n)\%n$

三、程序填空题 (Fill in the Blanks)

R5-1. *(9 分)* **MaxHeap Deletion** Please fill in the blanks in the program which deletes a given element at position `p` from a max-heap `H`.

```
Deletion ( PriorityQueue H,  int p )  /* delete the element H->Elements[p] */
{
   ElementType temp;
   int child;

   temp = H-> Elements[ H->Size-- ];
   if ( temp > H->Elements[p] ) {
      while ( (p != 1) && (temp > H->Elements[p/2]) ) {
         ________ (1) ________;
         p /= 2;
      }
   }
   else {
      while( (child = 2*p) <= H->Size) {
         if ( child != H->Size && ________ (2) ________ )
            child ++;
```

```
         if ( ________ (3) ________ ) {
            H->Elements[p] = H->Elements[child];
            p = child;
         }
         else
            break;
      }
   }
   H->Elements[p] = temp;
}
```

R5-2. *(9 分)* **Function Call Simulation with Linked Stack** There are n functions numbered from 1 to n. Function 1 is the main entry of the program.

- $cnt[i]$ represents the total number of child functions directly invoked by function i.
- $call[i][j]$ represents the ID of the j-th child function called by function i. All child functions are invoked strictly in the given order.

You need to use a linked stack with a head node to simulate the execution of nested function calls. Print `Enter x` when starting to execute function x, and print `Exit x` when function x finishes and exits.

```
typedef struct node {
    int data;
    struct node *next;
} StackNode, *Stack;

void Push(Stack S, int data) {
    StackNode *p = (StackNode*)malloc(sizeof(StackNode));
    p->data = data;
    ________ (1) ________;
    ________ (2) ________;
}

int Pop(Stack S) {
    StackNode *p = S->next;
    int data = p->data;
    S->next = p->next;
    free(p);
    return data;
}

int GetTop(Stack S) {
    return S->next->data;
```

```
}

void Simulate(int call[][MAXN], int cnt[]) {
    int ptr[MAXN] = {0};

    StackNode head;
    Stack S = &head;
    S->next = NULL;

    Push(S, 1);
    printf("Enter 1\n");

    while (________ (3) ________) {
        int u = GetTop(S);

        if (________ (4) ________) {
            int v = call[u][ptr[u]++];
            Push(S, v);
            printf("Enter %d\n", v);
        } else {
            Pop(S);
            printf("Exit %d\n", u);
        }
    }
}
```

参考答案

一、判断题

R1-1. T

R1-2. T

R1-3. T

R1-4. F

R1-5. T

R1-6. F

R1-7. F

R1-8. T

二、单选题

R2-1. C

R2-2. A

R2-3. C

R2-4. D

R2-5. B

R2-6. A

R2-7. A

R2-8. D

R2-9. C

R2-10. D

R2-11. D

R2-12. C

R2-13. D

三、程序填空题

R5-1 MaxHeap Deletion

(1) H−>Elements[p] = H−>Elements[p/2]

(2) H−>Elements[child+1]>H−>Elements[child]

(3) temp<H−>Elements[child]

R5-2 Function Call Simulation with Linked Stack

(1) p−>next = S−>next

(2) S−>next = p

(3) S−>next!=NULL

(4) cnt[u]−−